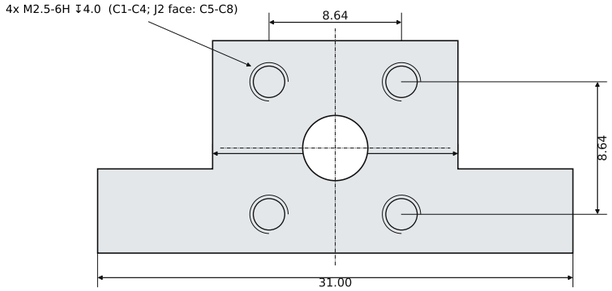
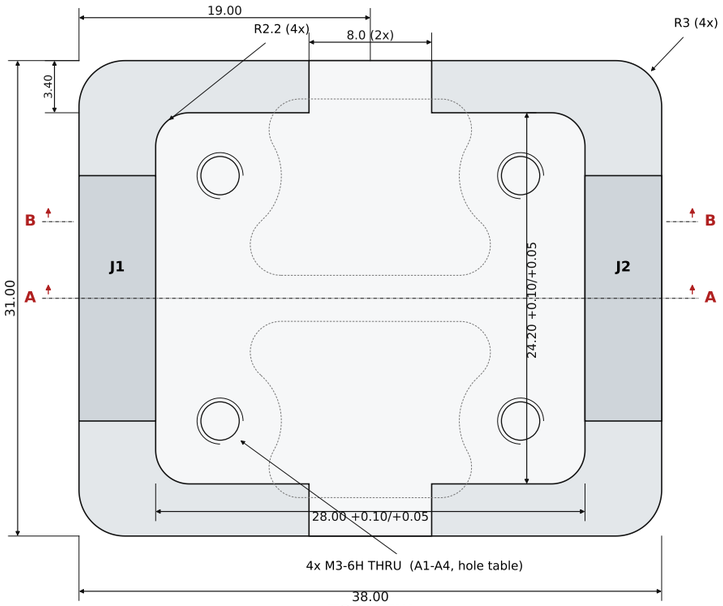


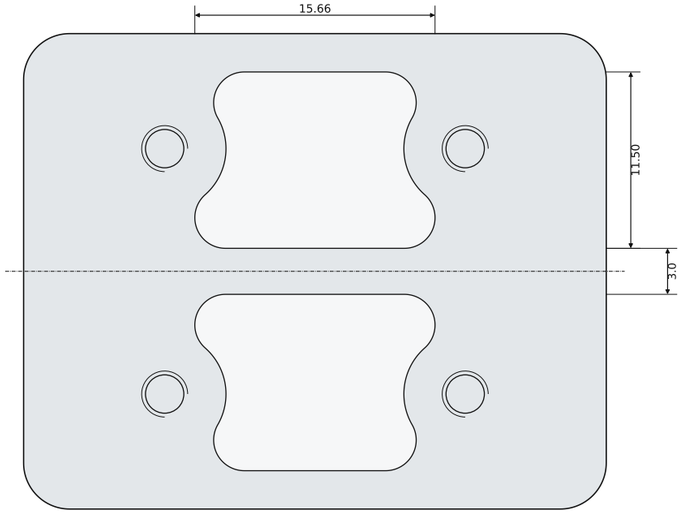
SECTION A-A (on the line axis)



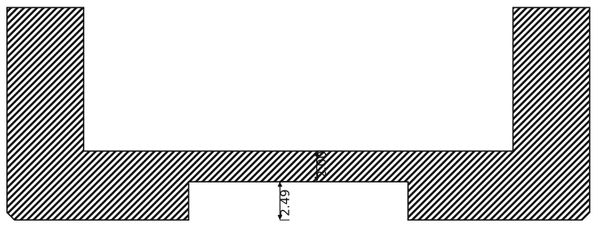
VIEW J1 (from -X)



TOP VIEW



VIEW FROM BELOW (pockets ↧2.49, R2 min)



SECTION B-B (Y 20.50)

HOLE TABLE (datum: plate SW corner, bottom face)

ID	Feature	X	Y	Z	Face
A1	M3-6H THRU (drill Ø2.5)	9.20	7.50	4.49	floor (-Z)
A2	M3-6H THRU (drill Ø2.5)	9.20	23.50	4.49	floor (-Z)
A3	M3-6H THRU (drill Ø2.5)	28.80	7.50	4.49	floor (-Z)
A4	M3-6H THRU (drill Ø2.5)	28.80	23.50	4.49	floor (-Z)
B1	Ø4.25 +0.05/0 THRU wall	0.00	15.50	6.85	J1 (-X)
C1	M2.5-6H ↧4.0 (drill Ø2.05 ↧4.6)	0.00	11.18	2.53	J1 (-X)
C2	M2.5-6H ↧4.0 (drill Ø2.05 ↧4.6)	0.00	11.18	11.17	J1 (-X)
C3	M2.5-6H ↧4.0 (drill Ø2.05 ↧4.6)	0.00	19.82	2.53	J1 (-X)
C4	M2.5-6H ↧4.0 (drill Ø2.05 ↧4.6)	0.00	19.82	11.17	J1 (-X)
B2	Ø4.25 +0.05/0 THRU wall	38.00	15.50	6.85	J2 (+X)
C5	M2.5-6H ↧4.0 (drill Ø2.05 ↧4.6)	38.00	11.18	2.53	J2 (+X)
C6	M2.5-6H ↧4.0 (drill Ø2.05 ↧4.6)	38.00	11.18	11.17	J2 (+X)
C7	M2.5-6H ↧4.0 (drill Ø2.05 ↧4.6)	38.00	19.82	2.53	J2 (+X)
C8	M2.5-6H ↧4.0 (drill Ø2.05 ↧4.6)	38.00	19.82	11.17	J2 (+X)

NOTES

1. Material: Aluminium EN AW-6061-T6, from plate.
2. Finish: chromate conversion MIL-DTL-5541 Type II Class 3, clear (conductive), all over, after machining.
3. Tolerances ISO 2768-mK unless stated; hole positions ±0.05.
4. Datum A = pocket floor, flatness 0.02. Bore axis B1/B2 to A 2.36 ±0.02; B1-B2 coaxial Ø0.05; wall inner faces ⊥ A 0.03.
5. Bore height is for a 1.73 mm cal PCB [ASSUMED]. Confirm the board thickness before machining: axis = A + t + 0.63.
6. Flange faces J1 / J2: flat 0.02, Ra 1.6, bores deburred.
7. Break sharp edges 0.2 max; bottom outside edges 0.5 x 45°.
8. Engrave 0.2 deep, Arial Bold: 'CRL-XB-CALP-15 REV A' on the -Y face (h 2.0), 'J1' / 'J2' on the wall tops (h 2.2).
9. Underside pockets: D4 end mill min. Mass 16.0 g.
10. CRL-XB-CALP-15_PLATE.step governs where not dimensioned.

BEL - CRL

ANTENNA / RFM LAB

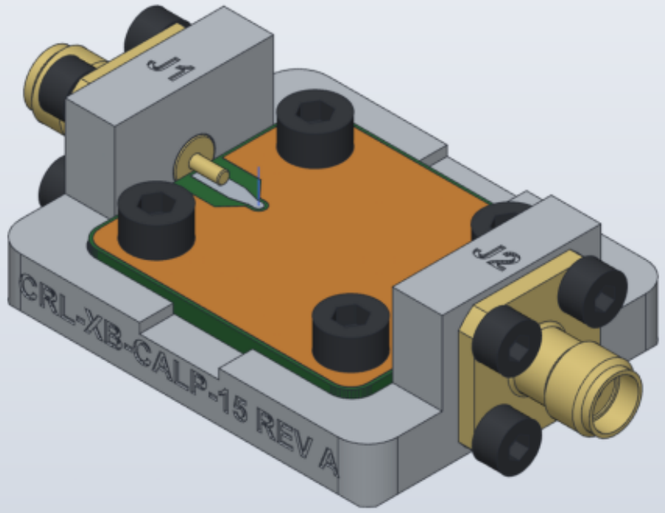
X-BAND MSL CAL LINE BASE PLATE

15 mm THRU (XBand_RO4350B_30mil_Thru_SHORT_15mm)

PLATE - CNC MACHINING

PART	DWG	REV	SHEET
CRL-XB-CALP-15	CRL-XB-CALP-15-DWG	A	1 / 2
DATE	UNITS	SCALE	PROJ.
06-11-2026	mm	2.5 : 1	1st angle

ASSEMBLY



EXPLODED

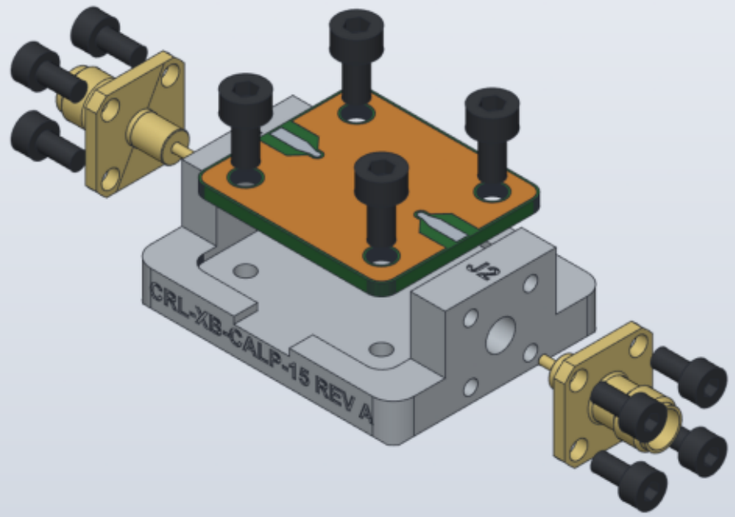
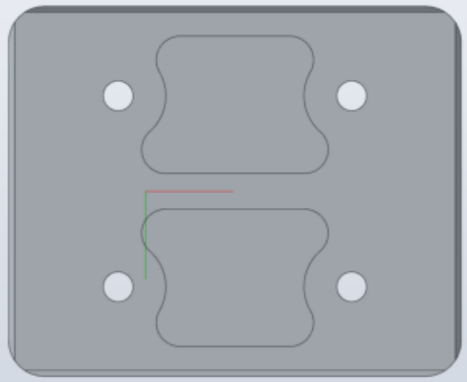


PLATE FROM BELOW



PARTS LIST

Item	Qty	Part	Description
1	1	CRL-XB-CALP-15_PLATE	Base plate, Aluminium EN AW-6061-T6 (sheet 1)
2	1	XBand_RO4350B_30mil_Thru_SHORT_15mm	Cal PCB (customer), MSL-SL-MSL THRU 15 mm
3	2	23_SMA-50-0-272/133_N	H+S SMA jack, 4-hole flange, pin on the launch pad
4	8	ISO 4762 M2.5 x 5	A2-70, flange screws, 0.4 Nm
5	4	ISO 4762 M3 x 6	A2-70, PCB to plate, 0.6 Nm
6	-	Sn62Pb36Ag2 / SAC305	Solder: SMA pin to the launch pad

ASSEMBLY

1. Clean the pocket floor and the PCB bottom ground (IPA). No paste/foil under the board.
2. Drop the PCB into the pocket (L1 up, J1 end = DXF x min); fit 4 x M3 x 6, finger tight.
3. Fit both SMA jacks through the walls (pin on the launch pad), 4 x M2.5 x 5 each, 0.4 Nm.
4. Push the PCB toward neither wall: the pocket centres it (0.11 gap each end); torque M3 to 0.6 Nm.
5. Solder each pin to its pad with a small fillet; check the pin is not lifted (gap < 0.05).
6. Check: DC continuity J1-J2 centre, no short to ground; then S-parameters 8-12 GHz.

BEL - CRL

ANTENNA / RFM LAB

X-BAND MSL CAL LINE BASE PLATE

15 mm THRU (XBand_RO4350B_30mil_Thru_SHORT_15mm)

ASSEMBLY + PARTS LIST

PART	DWG	REV	SHEET
CRL-XB-CALP-15	CRL-XB-CALP-15-DWG	A	2 / 2

DATE	UNITS	SCALE	PROJ.
06-11-2026	mm	NTS	1st angle